From the 2018 Administration

AP® CollegeBoard

AP Calculus BC

Practice Exam

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CALCULUS BC

SECTION I, Part A

Time—1 hour

Number of questions—30

NO CALCULATOR IS ALLOWED FOR THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in this exam booklet. Do not spend too much time on any one problem.

In this exam:

(1) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

(2) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix "arc" (e.g., $\sin^{-1} x = \arcsin x$).

1. If $y = x\sqrt{2x+5}$, then $y' =$

(A) $\dfrac{3x+5}{\sqrt{2x+5}}$

(B) $\dfrac{1}{\sqrt{2x+5}}$

(C) $\dfrac{1}{2\sqrt{2x+5}}$

(D) $\dfrac{5x+10}{2\sqrt{2x+5}}$

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2. $\int 2^x \, dx =$

(A) $2^x + C$ (B) $(\ln 2)2^x + C$ (C) $\dfrac{2^x}{\ln 2} + C$ (D) $\dfrac{2^{x+1}}{x+1} + C$

3. $\lim\limits_{x \to -7} \dfrac{x+7}{|x+7|}$ is

(A) -1 (B) 0 (C) 1 (D) nonexistent

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4. $\displaystyle\int \frac{\left(x^{1/3} - 4\right)^5}{6x^{2/3}}\,dx =$

(A) $\dfrac{\left(x^{1/3} - 4\right)^6}{12} + C$

(B) $\dfrac{\left(x^{1/3} - 4\right)^6}{6} + C$

(C) $\dfrac{5\left(x^{1/3} - 4\right)^4}{2} + C$

(D) $3\left(x^{1/3} - 4\right)^6 + C$

t (minutes)	0	5	10	15
$R(t)$ (people per minute)	100	100	75	55

5. During an evacuation drill, people leave a building at a rate of $R(t)$ people per minute, where t is the number of minutes since the start of the drill. Selected values of $R(t)$ are shown in the table above. Using a right Riemann sum with three subintervals and data from the table, what is the approximation of the number of people who leave the building during the first 15 minutes of the evacuation drill?

(A) 230 (B) 1150 (C) 1375 (D) 2075

6. What is the sum of the series $\frac{\pi}{e} - \frac{\pi}{e^2} + \frac{\pi}{e^3} - \frac{\pi}{e^4} + \cdots + (-1)^{n+1}\frac{\pi}{e^n} + \cdots$?

(A) $\frac{\pi}{e + \pi}$ (B) $\frac{\pi}{e + 1}$ (C) $\frac{\pi}{e - 1}$ (D) The series diverges.

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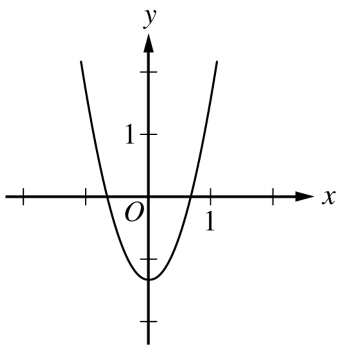
7. If $f(x) = \left(x^2 + 1\right)^3$, what is $\lim_{x \to -1} \frac{f(x) - f(-1)}{x + 1}$?

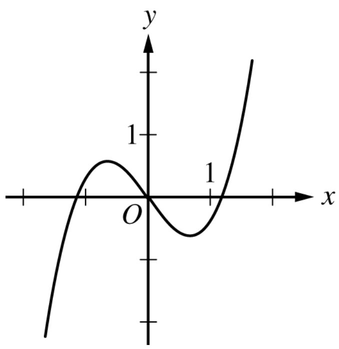
(A) -24 (B) -8 (C) 0 (D) 12



Graph of f''

8. The graph of f'', the second derivative of the function f, is shown above. Which of the following could be the graph of f ?

(A)



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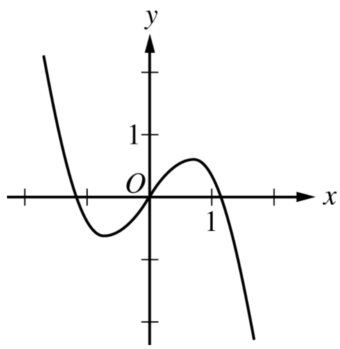
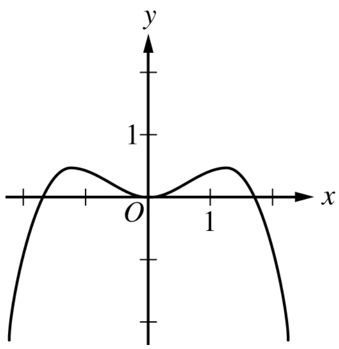
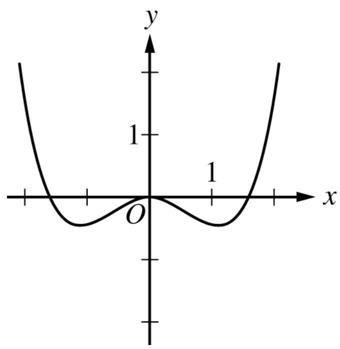


(C)



(D)



9. Let $y = f(x)$ be a differentiable function such that $\dfrac{dy}{dx} = \dfrac{x}{y}$ and $f(8) = 2$. What is the approximation of $f(8.1)$ using the line tangent to the graph of f at $x = 8$?

(A) 0.4 (B) 2.025 (C) 2.4 (D) 6

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10. The series $1 - x^2 + \dfrac{x^4}{2!} - \dfrac{x^6}{3!} + \dfrac{x^8}{4!} + \cdots + (-1)^n \dfrac{x^{2n}}{n!} + \cdots$ converges to which of the following?

(A) $\cos\left(x^2\right) + \sin\left(x^2\right)$ (B) $1 - x\sin x$ (C) $\cos x$ (D) e^{-x^2}

GO ON TO THE NEXT PAGE.

11. $\int x \cos(2x)\, dx =$

(A) $\frac{1}{2}x^2 \sin(2x) + C$

(B) $\frac{1}{2}x^2 \cos(2x) + \frac{1}{2}\sin(2x) + C$

(C) $\frac{1}{2}x\sin(2x) - \frac{1}{4}\cos(2x) + C$

(D) $\frac{1}{2}x\sin(2x) + \frac{1}{4}\cos(2x) + C$

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12. Given that $3x - \tan y = 4$, what is $\frac{dy}{dx}$ in terms of y ?

(A) $\frac{dy}{dx} = 3\sin^2 y$

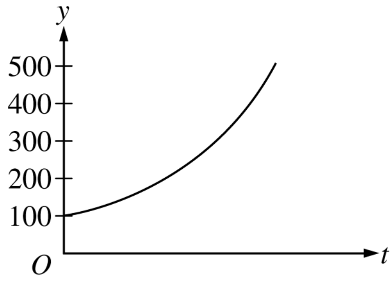
(B) $\frac{dy}{dx} = 3\cos^2 y$

(C) $\frac{dy}{dx} = 3\cos y \cot y$

(D) $\frac{dy}{dx} = \frac{3}{1 + 9y^2}$

13. Which of the following graphs is the solution to the logistic differential equation $\dfrac{dy}{dt} = \dfrac{y}{5}\left(1 - \dfrac{y}{500}\right)$ with the initial condition $y(0) = 100$?

(A)



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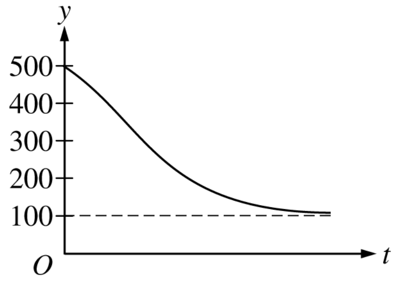
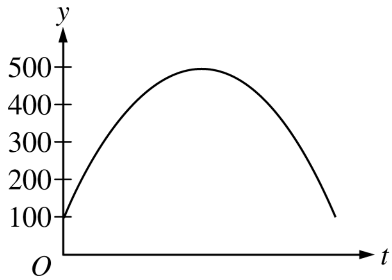
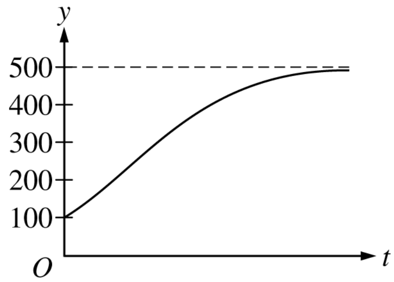


(C)

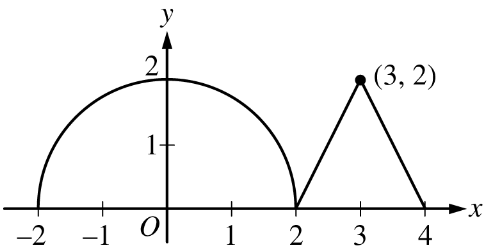


(D)



14. What is the absolute minimum value of $y = -\cos x - \sin x$ on the closed interval $\left[0, \dfrac{\pi}{2}\right]$?

(A) $-2\sqrt{2}$ (B) -2 (C) $-\sqrt{2}$ (D) -1



Graph of g'

15. The graph of g', the first derivative of the function g, consists of a semicircle of radius 2 and two line segments, as shown in the figure above. If $g(0) = 1$, what is $g(3)$?

(A) $\pi + 1$ (B) $\pi + 2$ (C) $2\pi + 1$ (D) $2\pi + 2$

16. $$\int \frac{1+3x}{(1-x)(3x-5)}\,dx =$$

(A) $2\ln|1-x| - 3\ln|3x-5| + C$

(B) $2\ln|1-x| - 27\ln|3x-5| + C$

(C) $-2\ln|1-x| - 3\ln|3x-5| + C$

(D) $-2\ln|1-x| - 9\ln|3x-5| + C$

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17. A spherical snowball is melting in such a way that it maintains its shape. The snowball is decreasing in volume at a constant rate of 8 cubic centimeters per hour. At what rate, in centimeters per hour, is the radius of the snowball decreasing at the instant when the radius is 10 centimeters? (The volume of a sphere of radius r is $V = \frac{4}{3}\pi r^3$.)

(A) $\frac{1}{50\pi}$ (B) $\frac{3}{50\pi}$ (C) 400π (D) 3200π

GO ON TO THE NEXT PAGE.

18. If $f(x) = \int_0^{x^3} \cos\left(t^2\right)\, dt$, then $f'(\sqrt{\pi}) =$

(A) $3\pi \sin\left(\pi^3\right)$ (B) $\cos\left(\pi^3\right)$ (C) $3\pi \cos \pi$ (D) $3\pi \cos\left(\pi^3\right)$

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19. What is the slope of the line tangent to the polar curve $r = 2\theta^2$ when $\theta = \pi$?

(A) 4π (B) $\frac{\pi}{2}$ (C) $\frac{2}{\pi}$ (D) $-2\pi^2$

20. Let g be a twice-differentiable, increasing function of t. If $g(0) = 20$ and $g(10) = 220$, which of the following must be true on the interval $0 < t < 10$?

(A) $g'(t) = 0$ for some t in the interval.

(B) $g'(t) = 20$ for some t in the interval.

(C) $g''(t) = 0$ for some t in the interval.

(D) $g''(t) > 0$ for all t in the interval.

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21. $$\int \frac{6x^2 - 4x - 25}{x - 2}\, dx =$$

(A) $3x^2 + 8x - 9\ln|x - 2| + C$

(B) $3x^2 + 8x + \dfrac{9}{(x - 2)^2} + C$

(C) $\left(2x^3 - 2x^2 - 25x\right)\ln|x - 2| + C$

(D) $\dfrac{2x^3 - 2x^2 - 25x}{\dfrac{x^2}{2} - 2x} + C$

GO ON TO THE NEXT PAGE.

22. Which of the following statements about convergence of the series $\sum_{n=1}^{\infty} \frac{1}{\ln(n+1)}$ is true?

(A) $\sum_{n=1}^{\infty} \frac{1}{\ln(n+1)}$ converges by comparison with $\sum_{n=1}^{\infty} \frac{1}{n}$.

(B) $\sum_{n=1}^{\infty} \frac{1}{\ln(n+1)}$ converges by comparison with $\sum_{n=1}^{\infty} \frac{1}{n^2}$.

(C) $\sum_{n=1}^{\infty} \frac{1}{\ln(n+1)}$ diverges by comparison with $\sum_{n=1}^{\infty} \frac{1}{n}$.

(D) $\sum_{n=1}^{\infty} \frac{1}{\ln(n+1)}$ diverges by comparison with $\sum_{n=1}^{\infty} \frac{1}{n^2}$.

GO ON TO THE NEXT PAGE.

23. The length of the curve $y = \sin x$ from $x = 0$ to $x = \dfrac{3\pi}{4}$ is given by

(A) $\int_0^{3\pi/4} \sin x \, dx$

(B) $\int_0^{3\pi/4} \sqrt{1 + \sin^2 x} \, dx$

(C) $\int_0^{3\pi/4} \sqrt{1 - \cos^2 x} \, dx$

(D) $\int_0^{3\pi/4} \sqrt{1 + \cos^2 x} \, dx$

GO ON TO THE NEXT PAGE.

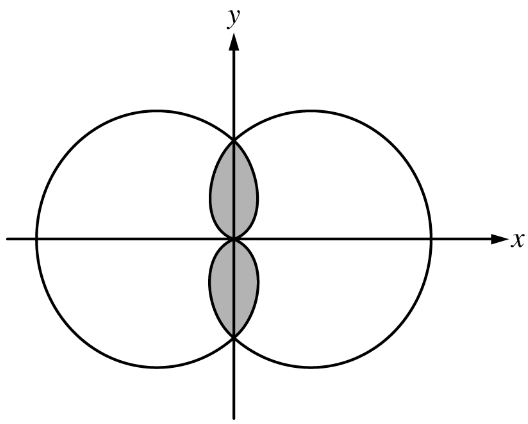
24. Let f be a function such that $f'(x) = \sin\left(x^2\right)$ and $f(0) = 0$. What are the first three nonzero terms of the Maclaurin series for f ?

(A) $x - \dfrac{x^5}{10} + \dfrac{x^9}{216}$

(B) $2x - x^5 + \dfrac{x^9}{12}$

(C) $\dfrac{x^3}{3} - \dfrac{x^7}{21} + \dfrac{x^{11}}{55}$

(D) $\dfrac{x^3}{3} - \dfrac{x^7}{42} + \dfrac{x^{11}}{1320}$



25. The polar curves $r = 1 - \cos\theta$ and $r = 1 + \cos\theta$ are shown in the figure above. Which of the following expressions gives the total area of the shaded regions?

(A) $\int_0^{\pi} (1 + \cos\theta)^2 \, d\theta$

(B) $\int_{\pi/2}^{\pi} (1 + \cos\theta)^2 \, d\theta$

(C) $2\int_0^{\pi/2} (1 - \cos\theta)^2 \, d\theta$

(D) $\int_0^{\pi/2} \left((1 - \cos\theta)^2 + (1 + \cos\theta)^2\right) d\theta$

GO ON TO THE NEXT PAGE.

x	10	11	12	13	14
$f(x)$	5	2	3	6	5

26. The table above gives values of the continuous function f at selected values of x. If f has exactly two critical points on the open interval $(10, 14)$, which of the following must be true?

(A) $f(x) > 0$ for all x in the open interval $(10, 14)$.

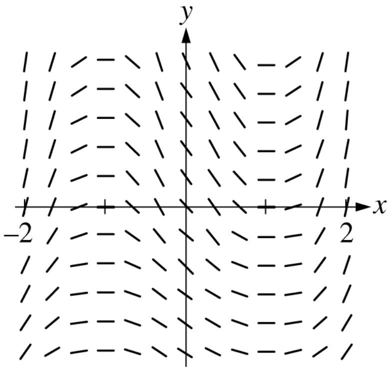
(B) $f'(x)$ exists for all x in the open interval $(10, 14)$.

(C) $f'(x) < 0$ for all x in the open interval $(10, 11)$.

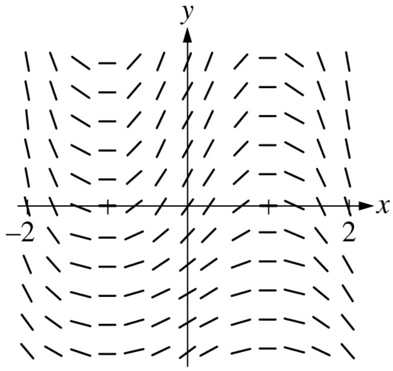
(D) $f'(12) \neq 0$

27. Let g be a function such that $g(y) > 0$ for all y. Which of the following could be a slope field for the differential equation $\frac{dy}{dx} = \left(x^2 - 1\right)g(y)$?

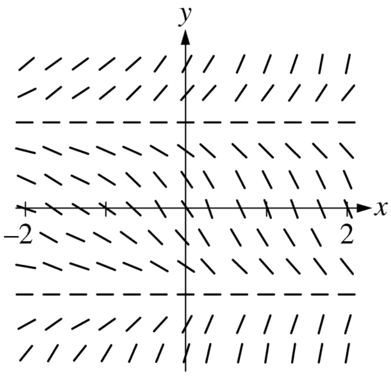
(A)



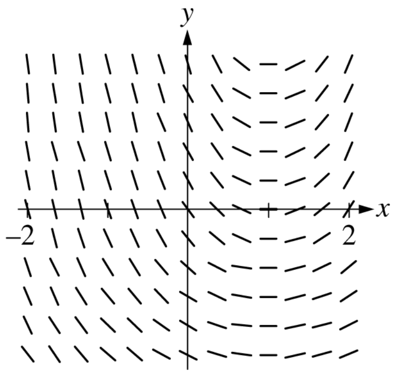
(B)



(C)



(D)



28. What are all values of p for which $\int_1^{\infty} \frac{1}{x^{3p+1}}\, dx$ converges?

(A) $p < 0$ (B) $p > -\frac{1}{3}$ (C) $p > 0$ (D) $p > 1$

29. The temperature of a solid at time $t \geq 0$ is modeled by the nonconstant function H and increases according to the differential equation $\frac{dH}{dt} = 2H + 1$, where $H(t)$ is measured in degrees Fahrenheit and t is measured in hours. Which of the following must be true?

(A) $H = H^2 + t + C$

(B) $\ln|2H + 1| = \frac{t}{2} + C$

(C) $\ln|2H + 1| = t + C$

(D) $\ln|2H + 1| = 2t + C$

30. If the power series $\sum_{n=0}^{\infty} a_n(x-4)^n$ converges at $x = 7$ and diverges at $x = 9$, which of the following must be true?

I. The series converges at $x = 1$.

II. The series converges at $x = 2$.

III. The series diverges at $x = -1$.

(A) I only

(B) II only

(C) I and II only

(D) II and III only

END OF PART A

IF YOU FINISH BEFORE TIME IS CALLED,
YOU MAY CHECK YOUR WORK ON PART A ONLY.

DO NOT GO ON TO PART B UNTIL YOU ARE TOLD TO DO SO.

CALCULUS BC

SECTION I, Part B

Time—45 minutes

Number of questions—15

A GRAPHING CALCULATOR IS REQUIRED FOR SOME QUESTIONS ON THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in this exam booklet. Do not spend too much time on any one problem.

BE SURE YOU ARE USING PAGE 3 OF THE ANSWER SHEET TO RECORD YOUR ANSWERS TO QUESTIONS NUMBERED 76–90.

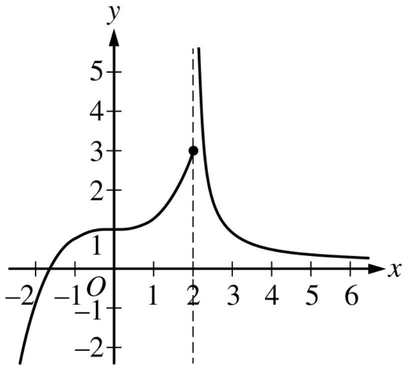
YOU MAY NOT RETURN TO PAGE 2 OF THE ANSWER SHEET.

In this exam:

(1) The exact numerical value of the correct answer does not always appear among the choices given. When this happens, select from among the choices the number that best approximates the exact numerical value.

(2) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

(3) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix "arc" (e.g., $\sin^{-1} x = \arcsin x$).



Graph of f

76. The graph of the function f is shown in the figure above. Which of the following statements must be false?

(A) $\lim_{x \to 2^-} f(x) = 3$

(B) $\lim_{x \to 2^+} f(x) = \infty$

(C) $\lim_{x \to 2} f(x) = f(2)$

(D) $\lim_{x \to \infty} f(x) = 0$

77. The rate at which water leaks from a tank, in gallons per hour, is modeled by R, a differentiable function of the number of hours after the leak is discovered. Which of the following is the best interpretation of $R'(3)$?

(A) The amount of water, in gallons, that has leaked out of the tank during the first three hours after the leak is discovered

(B) The amount of change, in gallons per hour, in the rate at which water is leaking during the three hours after the leak is discovered

(C) The rate at which water leaks from the tank, in gallons per hour, three hours after the leak is discovered

(D) The rate of change of the rate at which water leaks from the tank, in gallons per hour per hour, three hours after the leak is discovered

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78. On a certain day, the temperature, in degrees Fahrenheit, in a small town t hours after midnight $(t = 0)$ is modeled by the function $g(t) = 65 - 8\sin\left(\frac{\pi t}{12}\right)$. What is the average temperature in the town between 3 A.M. $(t = 3)$ and 6 A.M. $(t = 6)$, in degrees Fahrenheit?

(A) 57.609 (B) 57.797 (C) 58.172 (D) 59.907

GO ON TO THE NEXT PAGE.

x	0.0	0.5	1.0	1.5	2.0
$f'(x)$	1.0	0.7	0.5	0.4	0.3

79. Let $y = f(x)$ be the solution to the differential equation $\frac{dy}{dx} = f'(x)$ with initial condition $f(1) = 5$. Selected values of $f'(x)$ are given in the table above. What is the approximation for $f(2)$ if Euler's method is used with a step size of 0.5, starting at $x = 1$?

(A) 5.35 (B) 5.45 (C) 5.50 (D) 5.90

GO ON TO THE NEXT PAGE.

80. The first derivative of the function f is defined by $f'(x) = \left(x^2 + 1\right)\sin(3x - 1)$ for $-1.5 < x < 1.5$. On which of the following intervals is the graph of f concave up?

(A) $(-1.5, -1.341)$ and $(-0.240, 0.964)$

(B) $(-1.341, -0.240)$ and $(0.964, 1.5)$

(C) $(-0.714, 0.333)$ and $(1.381, 1.5)$

(D) $(-1.5, -0.714)$ and $(0.333, 1.381)$

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81. For $t \geq 0$, the velocity of a particle moving along the x-axis is given by $v(t) = t^3 - 6t^2 + 10t - 4$. At what time t does the direction of motion of the particle change from right to left?

(A) 0.586 (B) 1.184 (C) 2.000 (D) 2.816

82. Let f be a function such that $f(1) = -2$ and $f(5) = 7$. Which of the following conditions ensures that $f(c) = 0$ for some value c in the open interval $(1, 5)$?

(A) $\int_1^5 f(x)\,dx$ exists.

(B) f is increasing on the closed interval [1, 5].

(C) f is continuous on the closed interval [1, 5].

(D) f is defined for all values of x in the closed interval [1, 5].

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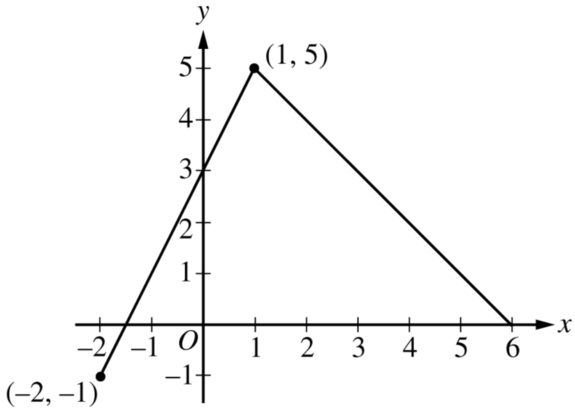
83. For time $t > 0$, the position of an object moving in the xy-plane is given by the parametric equations $x(t) = t\cos\left(\frac{t}{2}\right)$ and $y(t) = \sqrt{t^2 + 2t}$. What is the speed of the object at time $t = 1$?

(A) 1.155 (B) 1.319 (C) 1.339 (D) 1.810



Graph of g

84. The graph of the function g is shown above. If f is the function given by $f(x) = g(g(x))$, what is the value of $f'(0)$?

(A) -2 (B) -1 (C) 2 (D) 3

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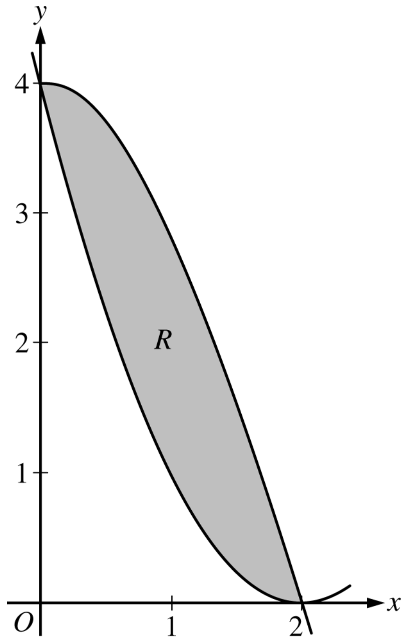
85. Let f be a function such that $f(-x) = -f(x)$ for all x. If $\int_0^2 f(x)\,dx = 5$, then $\int_{-2}^{2} (f(x) + 6)\,dx =$

(A) 6 (B) 16 (C) 24 (D) 34



86. Let R be the region in the first quadrant bounded by the graphs of $y = 4\cos\left(\frac{\pi x}{4}\right)$ and $y = (x - 2)^2$, as shown in the figure above. The region R is the base of a solid. For the solid, each cross section perpendicular to the x-axis is an isosceles right triangle with a leg in region R. What is the volume of the solid?

(A) 1.775 (B) 3.549 (C) 4.800 (D) 5.575

GO ON TO THE NEXT PAGE.

87. Suppose $\lim_{n\to\infty} a_n = \infty$ and $a_{n+1} \geq a_n > 0$ for all $n \geq 1$. Which of the following statements must be true?

(A) $\sum_{n=1}^{\infty} \frac{1}{a_n}$ diverges.

(B) $\sum_{n=1}^{\infty} (-1)^n a_n$ converges.

(C) $\sum_{n=1}^{\infty} \frac{1}{a_n}$ converges.

(D) $\sum_{n=1}^{\infty} \frac{(-1)^n}{a_n}$ converges.

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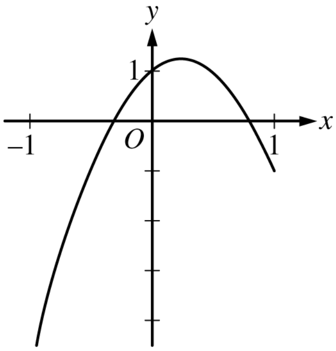
88. At time $t \geq 0$, a particle moving in the xy-plane has velocity vector given by $v(t) = \left\langle 3, 2^{-t^2} \right\rangle$. If the particle is at the point $\left(1, \frac{1}{2}\right)$ at time $t = 0$, how far is the particle from the origin at time $t = 1$?

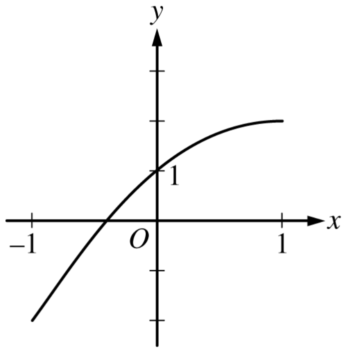
(A) 2.304 (B) 3.107 (C) 4.209 (D) 5.310

89. Let f be a function with $f(0) = 1$, $f'(0) = 2$, and $f''(0) = -2$. Which of the following could be the graph of the second-degree Taylor polynomial for f about $x = 0$?

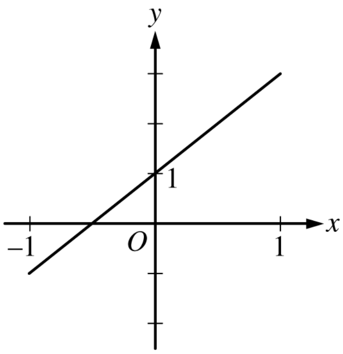
(A)



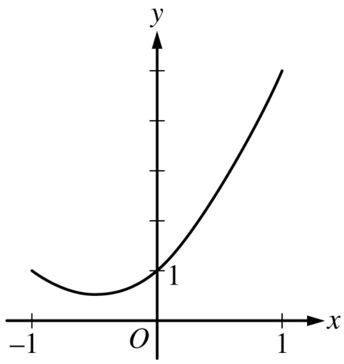
(B)



(C)



(D)



90. The series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$ converges to S. Based on the alternating series error bound, what is the least number of terms in the series that must be summed to guarantee a partial sum that is within 0.03 of S ?

(A) 34 (B) 333 (C) 1111 (D) 9999

CALCULUS BC

SECTION II, Part A

Time—30 minutes

Number of questions—2

A GRAPHING CALCULATOR IS REQUIRED FOR THESE QUESTIONS.

t (minutes)	0	1	5	6	8
$g(t)$ (cubic feet per minute)	12.8	15.1	20.5	18.3	22.7

1. Grain is being added to a silo. At time $t = 0$, the silo is empty. The rate at which grain is being added is modeled by the differentiable function g, where $g(t)$ is measured in cubic feet per minute for $0 \le t \le 8$ minutes. Selected values of $g(t)$ are given in the table above.

(a) Using the data in the table, approximate $g'(3)$. Using correct units, interpret the meaning of $g'(3)$ in the context of the problem.

(b) Write an integral expression that represents the total amount of grain added to the silo from time $t = 0$ to time $t = 8$. Use a right Riemann sum with the four subintervals indicated by the data in the table to approximate the integral.

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Continue question 1 on page 5.

(c) The grain in the silo is spoiling at a rate modeled by $w(t) = 32 \cdot \sqrt{\sin\left(\frac{\pi t}{74}\right)}$, where $w(t)$ is measured in cubic feet per minute for $0 \le t \le 8$ minutes. Using the result from part (b), approximate the amount of unspoiled grain remaining in the silo at time $t = 8$.

(d) Based on the model in part (c), is the amount of unspoiled grain in the silo increasing or decreasing at time $t = 6$? Show the work that leads to your answer.

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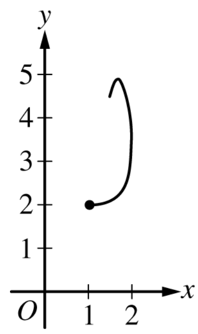

2. A particle moving in the xy-plane has position $(x(t), y(t))$ at time $t \geq 0$, where $\frac{dx}{dt} = \cos\left(t^2\right)$ and $\frac{dy}{dt} = e^t \sin\left(t^2\right)$. At time $t = 0$, the particle is at position $(1, 2)$. The figure above shows the path of the particle for $0 \leq t \leq 2$.

(a) Find the position of the particle at time $t = 2$.

(b) Find the slope of the line tangent to the particle's path at time $t = 2$.

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Continue question 2 on page 7.

(c) Find the speed of the particle at time $t = 2$. Find the acceleration vector of the particle at time $t = 2$.

(d) Consider a rectangle with vertices at points $(0, 0)$, $(x(t), 0)$, $(x(t), y(t))$, and $(0, y(t))$ at time $t \geq 0$. For $0 \leq t \leq 2$, at what time t is the perimeter of the rectangle a maximum? Justify your answer.

CALCULUS BC

SECTION II, Part B

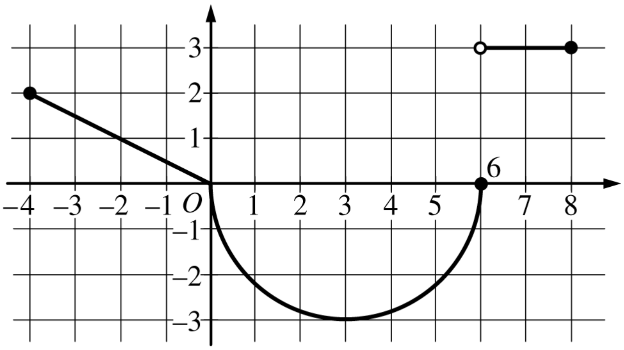
Time—1 hour

Number of questions—4

NO CALCULATOR IS ALLOWED FOR THESE QUESTIONS.

DO NOT BREAK THE SEALS UNTIL YOU ARE TOLD TO DO SO.

NO CALCULATOR ALLOWED



Graph of g

3. The function g is defined on the closed interval $[-4, 8]$. The graph of g consists of two linear pieces and a semicircle, as shown in the figure above. Let f be the function defined by $f(x) = 3x + \int_0^x g(t)\, dt$.

(a) Find $f(7)$ and $f'(7)$.

(b) Find the value of x in the closed interval $[-4, 3]$ at which f attains its maximum value. Justify your answer.

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Continue question 3 on page 15.

NO CALCULATOR ALLOWED

(c) For each of $\lim_{x\to 0^-} g'(x)$ and $\lim_{x\to 0^+} g'(x)$, find the value or state that it does not exist.

(d) Find $\lim_{x\to -2} \dfrac{f(x)+7}{e^{3x+6}-1}$.

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NO CALCULATOR ALLOWED

4. Let g be the function that satisfies $g(0) = 0$ and whose derivative satisfies $g'(x) = 2|x|$.

(a) Find expressions for $g(x)$ and $g''(x)$.

(b) Find the x-coordinate, if any, of each point of inflection of the graph of $y = g(x)$. Explain your reasoning.

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Continue question 4 on page 17.

NO CALCULATOR ALLOWED

(c) Let $h(x) = \int_0^x \sqrt{1 + 4t^2}\, dt$. For $x \geq 0$, $h(x)$ is the length of the graph of g from $t = 0$ to $t = x$. Use Euler's method, starting at $x = 0$ with two steps of equal size, to approximate $h(4)$.

(d) Find the value of $\int_{\pi/2}^{\pi} g'(x) \cos x\, dx$.

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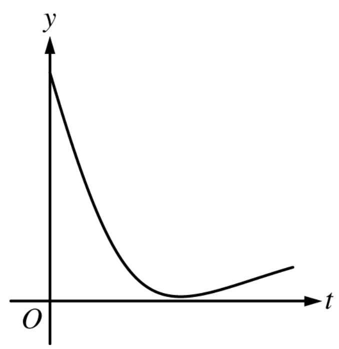
NO CALCULATOR ALLOWED

5. During a chemical reaction, the function $y = f(t)$ models the amount of a substance present, in grams, at time t seconds. At the start of the reaction $(t = 0)$, there are 10 grams of the substance present. The function $y = f(t)$ satisfies the differential equation $\frac{dy}{dt} = -0.02y^2$.

(a) Use the line tangent to the graph of $y = f(t)$ at $t = 0$ to approximate the amount of the substance remaining at time $t = 2$ seconds.

(b) Using the given differential equation, determine whether the graph of f could resemble the following graph. Give a reason for your answer.



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Continue question 5 on page 19.

NO CALCULATOR ALLOWED

(c) Find an expression for $y = f(t)$ by solving the differential equation $\dfrac{dy}{dt} = -0.02y^2$ with the initial condition $f(0) = 10$.

(d) Determine whether the amount of the substance is changing at an increasing or a decreasing rate. Explain your reasoning.

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NO CALCULATOR ALLOWED

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$	$f^{(4)}(x)$
0	4	5	–1	$-\frac{15}{2}$	23
1	8	3	–2	$\frac{3}{2}$	$\frac{2}{5}$

6. Let f be a function having derivatives of all orders for all real numbers. Selected values of f and its first four derivatives are shown in the table above.

(a) Write the second-degree Taylor polynomial for f about $x = 0$ and use it to approximate $f(0.2)$.

(b) Let g be a function such that $g(x) = f\left(x^3\right)$. Write the fifth-degree Taylor polynomial for g', the derivative of g, about $x = 0$.

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Continue question 6 on page 21.

NO CALCULATOR ALLOWED

(c) Write the third-degree Taylor polynomial for f about $x = 1$.

(d) It is known that $\left| f^{(4)}(x) \right| \le 300$ for $0 \le x \le 1.125$. The third-degree Taylor polynomial for f about $x = 1$, found in part (c), is used to approximate $f(1.1)$. Use the Lagrange error bound along with the information about $f^{(4)}(x)$ to find an upper bound on the error of the approximation.

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Answer Key for AP Calculus BC
Practice Exam, Section I

Question 1: A
Question 2: C
Question 3: D
Question 4: A
Question 5: B
Question 6: B
Question 7: A
Question 8: D
Question 9: C
Question 10: D
Question 11: D
Question 12: B
Question 13: D
Question 14: C
Question 15: B
Question 16: A
Question 17: A
Question 18: D
Question 19: B
Question 20: B
Question 21: A
Question 22: C
Question 23: D
Question 24: D
Question 25: C
Question 26: D
Question 27: A
Question 28: C
Question 29: D
Question 30: B
Question 76: C
Question 77: D
Question 78: B
Question 79: B
Question 80: A
Question 81: C
Question 82: C
Question 83: B
Question 84: A
Question 85: C
Question 86: A
Question 87: D
Question 88: C
Question 89: B
Question 90: C

2018 AP Calculus BC
Question Descriptors and Performance Data

Multiple-Choice Questions

Question	Learning Objective	Essential Knowledge	Mathematical Practice for AP Calculus 1	Mathematical Practice for AP Calculus 2	Key	% Correct
1	2.1C	2.1C3	Implementing algebraic/ computational processes	Building notational fluency	A	67
2	3.3B(a)	3.3B3	Implementing algebraic/ computational processes	Building notational fluency	C	58
3	1.1A(b)	1.1A3	Reasoning with definitions and theorems	Building notational fluency	D	35
4	3.3B(a)	3.3B5	Implementing algebraic/ computational processes	Building notational fluency	A	70
5	3.2B	3.2B2	Implementing algebraic/ computational processes	Connecting multiple representations	B	89
6	4.1B	4.1B1	Implementing algebraic/ computational processes	Building notational fluency	B	43
7	2.1A	2.1A2	Reasoning with definitions and theorems	Implementing algebraic/ computational processes	A	62
8	2.2A	2.2A3	Connecting concepts	Connecting multiple representations	D	72
9	2.3B	2.3B2	Connecting concepts	Implementing algebraic/ computational processes	C	83
10	4.2B	4.2B2	Implementing algebraic/ computational processes	Building notational fluency	D	51
11	3.3B(a)	3.3B5	Implementing algebraic/ computational processes	Building notational fluency	D	65
12	2.1C	2.1C5	Implementing algebraic/ computational processes	Connecting concepts	B	69
13	3.5A	3.5A1	Connecting multiple representations	Connecting concepts	D	86
14	2.2A	2.2A1	Implementing algebraic/ computational processes	Connecting concepts	C	64
15	3.2C	3.2C1	Connecting multiple representations	Implementing algebraic/ computational processes	B	66
16	3.3B(a)	3.3B5	Implementing algebraic/ computational processes	Building notational fluency	A	31
17	2.3C	2.3C2	Implementing algebraic/ computational processes	Connecting concepts	A	80
18	3.3A	3.3A2	Connecting concepts	Implementing algebraic/ computational processes	D	59
19	2.1C	2.1C7	Implementing algebraic/ computational processes	Connecting concepts	B	46
20	2.4A	2.4A1	Reasoning with definitions and theorems	Connecting concepts	B	67
21	3.3B(a)	3.3B5	Implementing algebraic/ computational processes	Building notational fluency	A	51
22	4.1A	4.1A6	Reasoning with definitions and theorems	Implementing algebraic/ computational processes	C	64
23	3.4D	3.4D3	Connecting concepts	Reasoning with definitions and theorems	D	81
24	4.2B	4.2B5	Implementing algebraic/ computational processes	Building notational fluency	D	40
25	3.4D	3.4D1	Connecting multiple representations	Reasoning with definitions and theorems	C	55
26	2.2A	2.2A1	Connecting multiple representations	Reasoning with definitions and theorems	D	36
27	2.3F	2.3F1	Connecting multiple representations	Connecting concepts	A	73
28	3.2D	3.2D2	Connecting concepts	Reasoning with definitions and theorems	C	68
29	3.5A	3.5A2	Connecting concepts	Implementing algebraic/ computational processes	D	41
30	4.2C	4.2C1	Reasoning with definitions and theorems	Connecting concepts	B	17

2018 AP Calculus BC
Question Descriptors and Performance Data

Question	Learning Objective	Essential Knowledge	Mathematical Practice for AP Calculus 1	Mathematical Practice for AP Calculus 2	Key	% Correct
76	1.2A	1.2A1	Connecting multiple representations	Reasoning with definitions and theorems	C	91
77	2.3A	2.3A2	Connecting concepts	Building notational fluency	D	81
78	3.4B	3.4B1	Implementing algebraic/ computational processes	Connecting concepts	B	85
79	2.3F	2.3F2	Implementing algebraic/ computational processes	Connecting multiple representations	B	83
80	2.2A	2.2A1	Connecting concepts	Implementing algebraic/ computational processes	A	71
81	2.3C	2.3C1	Implementing algebraic/ computational processes	Connecting concepts	C	76
82	1.2B	1.2B1	Reasoning with definitions and theorems	Building notational fluency	C	83
83	2.3C	2.3C4	Reasoning with definitions and theorems	Implementing algebraic/ computational processes	B	71
84	2.1C	2.1C4	Connecting multiple representations	Implementing algebraic/ computational processes	A	71
85	3.2C	3.2C2	Reasoning with definitions and theorems	Implementing algebraic/ computational processes	C	46
86	3.4D	3.4D2	Implementing algebraic/ computational processes	Connecting multiple representations	A	54
87	4.1A	4.1A6	Reasoning with definitions and theorems	Building notational fluency	D	52
88	3.4C	3.4C2	Implementing algebraic/ computational processes	Connecting concepts	C	61
89	4.2A	4.2A1	Connecting multiple representations	Connecting concepts	B	52
90	4.1B	4.1B2	Reasoning with definitions and theorems	Implementing algebraic/ computational processes	C	62

2018 AP Calculus BC
Question Descriptors and Performance Data

Free-Response Questions

Question	Learning Objective	Essential Knowledge	Mathematical Practice for AP Calculus	Mean
1	2.1B|2.3A|2.3A|3.2B|3.3B(b)|3.4A	2.1B1|2.3A1|2.3A2|3.2B2|3.3B2|3.4A2	Reasoning with definitions and theorems|Connecting concepts|Implementing algebraic/computational processes|Connecting multiple representations|Building notational fluency|Communicating	6.23
2	1.2B|2.1C|2.3C|2.3C|3.4C	1.2B1|2.1C7|2.3C3|2.3C4|3.4C2	Reasoning with definitions and theorems|Connecting concepts|Implementing algebraic/computational processes|Connecting multiple representations|Building notational fluency|Communicating	4.67
3	1.1A(b)|1.1B|1.1C|2.2A|3.2C|3.2C|3.3A	1.1A3|1.1B1|1.1C3|2.2A1|3.2C1|3.2C3|3.3A2	Reasoning with definitions and theorems|Connecting concepts|Implementing algebraic/computational processes|Connecting multiple representations|Building notational fluency|Communicating	4.19
4	2.1C|2.2A|2.2B|2.3F|3.3A|3.3B(b)	2.1C2|2.2A1|2.2B1|2.3F2|3.3A2|3.3B5	Reasoning with definitions and theorems|Connecting concepts|Implementing algebraic/computational processes|Building notational fluency|Communicating	3.22
5	2.1C|2.1D|2.2A|2.3B|3.5A	2.1C4|2.1D1|2.2A1|2.3B2|3.5A2	Reasoning with definitions and theorems|Connecting concepts|Implementing algebraic/computational processes|Connecting multiple representations|Building notational fluency|Communicating	3.95
6	4.2A|4.2A|4.2A|4.2B	4.2A1|4.2A2|4.2A4|4.2B5	Reasoning with definitions and theorems|Connecting concepts|Implementing algebraic/computational processes|Connecting multiple representations|Building notational fluency|Communicating	4.00

Question 1

(a) $g'(3) \approx \dfrac{g(5)-g(1)}{5-1} = \dfrac{20.5-15.1}{4} = 1.35$ At time $t = 3$ minutes, the rate at which grain is being added to the silo is increasing at a rate of 1.35 cubic feet per minute per minute.	$2 : \begin{cases} 1 : \text{approximation} \\ 1 : \text{interpretation with units} \end{cases}$
(b) The total amount of grain added to the silo from time $t = 0$ to time $t = 8$ is $\int_0^8 g(t)\,dt$ cubic feet. $\int_0^8 g(t)\,dt \approx g(1)\cdot(1-0) + g(5)\cdot(5-1) + g(6)\cdot(6-5) + g(8)\cdot(8-6)$ $= 15.1\cdot 1 + 20.5\cdot 4 + 18.3\cdot 1 + 22.7\cdot 2 = 160.8$	$3 : \begin{cases} 1 : \text{integral expression} \\ 1 : \text{right Riemann sum} \\ 1 : \text{approximation} \end{cases}$
(c) $\int_0^8 w(t)\,dt = 99.051497$ The approximate amount of unspoiled grain remaining in the silo at time $t = 8$ is $160.8 - \int_0^8 w(t)\,dt = 61.749$ (or 61.748) cubic feet.	$2 : \begin{cases} 1 : \text{integral} \\ 1 : \text{answer} \end{cases}$
(d) $g(6) - w(6) = 18.3 - 16.063173 = 2.236827 > 0$ Because $g(6) - w(6) > 0$, the amount of unspoiled grain is increasing at time $t = 6$.	$2 : \begin{cases} 1 : \text{considers } g(6) - w(6) \\ 1 : \text{answer} \end{cases}$

Question 2

(a) $x(2) = 1 + \int_0^2 \frac{dx}{dt}\,dt = 1.461461$

$y(2) = 2 + \int_0^2 \frac{dy}{dt}\,dt = 4.268236$

The position of the particle at time $t = 2$ is $(1.461, 4.268)$.

$3 : \begin{cases} 1 : \text{integrals} \\ 1 : \text{uses initial condition} \\ 1 : \text{position} \end{cases}$

(b) $\left.\frac{dy}{dx}\right|_{t=2} = \left.\frac{dy/dt}{dx/dt}\right|_{t=2} = 8.555206$

The slope of the line tangent to the particle's path at time $t = 2$ is 8.555.

1 : slope

(c) $\left.\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\right|_{t=2} = 5.630128$

The speed of the particle at time $t = 2$ is 5.630.

$x''(2) = 3.027210$
$y''(2) = -24.911294$

The acceleration vector of the particle at time $t = 2$ is $\langle 3.027, -24.911 \rangle$.

$2 : \begin{cases} 1 : \text{speed} \\ 1 : \text{acceleration vector} \end{cases}$

(d) Perimeter $= p(t) = 2(x(t) + y(t))$

$p'(t) = 2\left(\frac{dx}{dt} + \frac{dy}{dt}\right) = 0 \Rightarrow t = 1.721837$

Because $t = 1.721837$ is the only interior critical point for $0 \le t \le 2$ and $p'(t)$ changes from positive to negative at this point, the perimeter of the rectangle is a maximum at $t = 1.722$ (or $t = 1.721$).

$3 : \begin{cases} 1 : \text{sets } p'(t) = 0 \\ 2 : \text{answer with justification} \end{cases}$

Question 3

(a) $f(7) = 3 \cdot 7 + \int_0^7 g(t)\,dt = 21 - \frac{9\pi}{2} + 3 = 24 - \frac{9\pi}{2}$ $f'(7) = 3 + g(7) = 3 + 3 = 6$	$2 : \begin{cases} 1 : f(7) \\ 1 : f'(7) \end{cases}$
(b) On the interval $-4 \le x \le 3$, $f'(x) = 3 + g(x)$. Because $f'(x) \ge 0$ for $-4 \le x \le 3$, f is nondecreasing over the entire interval, and the maximum must occur when $x = 3$.	2 : answer with justification
(c) $\lim_{x \to 0^-} g'(x) = -\frac{1}{2}$ $\lim_{x \to 0^+} g'(x)$ does not exist.	$2 : \begin{cases} 1 : \text{left-hand limit} \\ 1 : \text{right-hand limit} \end{cases}$
(d) $\lim_{x \to -2} (f(x) + 7) = -6 + \int_0^{-2} g(t)\,dt + 7 = 0$ $\lim_{x \to -2} \left(e^{3x+6} - 1\right) = 0$ Using L'Hospital's Rule, $\lim_{x \to -2} \frac{f(x) + 7}{e^{3x+6} - 1} = \lim_{x \to -2} \frac{f'(x)}{3e^{3x+6}} = \frac{3 + g(-2)}{3} = \frac{3 + 1}{3} = \frac{4}{3}.$	$3 : \begin{cases} 1 : \text{limits equal 0} \\ 1 : \text{applies L'Hospital's Rule} \\ 1 : \text{answer} \end{cases}$ Note: max $1/3$ [1-0-0] if no limit notation attached to a ratio of derivatives

Question 4

(a) $g(x) = \begin{cases} -x^2 & \text{for } x < 0 \\ 0 & \text{for } x = 0 \\ x^2 & \text{for } x > 0 \end{cases}$

$g''(x) = \begin{cases} -2 & \text{for } x < 0 \\ 2 & \text{for } x > 0 \end{cases}$

$2 : \begin{cases} 1 : g(x) \\ 1 : g''(x) \end{cases}$

(b) The only point of inflection of the graph of $y = g(x)$ occurs at $x = 0$, where $g''(x)$ changes sign from negative to positive.

1 : answer with explanation

(c) $h(x) = \int_0^x \sqrt{1 + 4t^2}\, dt \Rightarrow h'(x) = \sqrt{1 + 4x^2}$

$h(0) = 0$
$h(2) \approx h(0) + 2 \cdot h'(0) = 0 + 2 \cdot 1 = 2$
$h(4) \approx h(2) + 2 \cdot h'(2) \approx 2 + 2\sqrt{17}$

$3 : \begin{cases} 1 : \text{Fundamental Theorem of Calculus} \\ 1 : \text{Euler's method with two steps} \\ 1 : \text{approximation} \end{cases}$

(d) $g'(x) = 2x$ for $x > 0$

$u = 2x \qquad dv = \cos x\, dx$
$du = 2\, dx \qquad v = \sin x$

$$\begin{aligned} \int_{\pi/2}^{\pi} 2x \cos x\, dx &= 2x \sin x \Big|_{\pi/2}^{\pi} - \int_{\pi/2}^{\pi} 2 \sin x\, dx \\ &= 2x \sin x \Big|_{\pi/2}^{\pi} + 2 \cos x \Big|_{\pi/2}^{\pi} \\ &= 0 - \pi + (-2) - 0 \\ &= -\pi - 2 \end{aligned}$$

$3 : \begin{cases} 2 : \text{integration by parts} \\ 1 : \text{answer} \end{cases}$

Question 5

(a) $y'(0) = -0.02\left(10^2\right) = -2$

An equation for the line tangent to the graph of $y = f(t)$ at $t = 0$ is $y = 10 - 2t$.

$y(2) \approx 10 - 2(2) = 6$ grams

$2 : \begin{cases} 1 : y'(0) \\ 1 : \text{approximation} \end{cases}$

(b) $\dfrac{dy}{dt} = -0.02y^2 \le 0,$ so the graph of f is nonincreasing.

The graph of f cannot resemble the given graph because the given graph is increasing on a portion of its domain.

1 : answer with reason

(c) $\displaystyle\int\left(-\frac{1}{y^2}\right) dy = \int 0.02\ dt$

$\dfrac{1}{y} = 0.02t + C$

$\dfrac{1}{10} = 0.02(0) + C \Rightarrow C = 0.1$

$\dfrac{1}{y} = 0.02t + 0.1 \Rightarrow y = \dfrac{1}{0.02t + 0.1} = \dfrac{50}{t+5}$

Note: this solution is valid for $t > -5$.

$4 : \begin{cases} 1 : \text{separation of variables} \\ 1 : \text{antiderivatives} \\ 1 : \text{constant of integration} \\ \quad \text{and uses initial condition} \\ 1 : \text{answer} \end{cases}$

Note: max $2/4$ [1-1-0-0] if no constant of integration

Note: $0/4$ if no separation of variables

(d) $$\frac{d^2y}{dt^2} = -0.04y\frac{dy}{dt} = -0.04y\left(-0.02y^2\right) = 0.0008y^3$$

Because $y > 0$, $0.0008y^3 > 0$.
The amount of substance is changing at an increasing rate.

— OR —

From part (c), $f(t) = \dfrac{50}{t+5}$, and from context, $t \ge 0$.

$f'(t) = \dfrac{-50}{(t+5)^2}$ and $f''(t) = \dfrac{100}{(t+5)^3} > 0$ for $t \ge 0$.

The amount of substance is changing at an increasing rate.

$2 : \begin{cases} 1 : \dfrac{d^2y}{dt^2} \\ 1 : \text{answer with reason} \end{cases}$

Question 6

(a) The second-degree Taylor polynomial for f about $x=0$ is $4+5x-\frac{1}{2}x^2$. $f(0.2) \approx 4+5(0.2)-\frac{1}{2}(0.04)=4.98$	$3: \begin{cases} 2: \text{second-degree} \\ \quad\ \text{Taylor polynomial} \\ 1: \text{approximation} \end{cases}$
(b) The fifth-degree Taylor polynomial for g' about $x=0$ is $\frac{d}{dx}\left(4+5\left(x^3\right)-\frac{1}{2}\left(x^3\right)^2\right)=15x^2-3x^5.$	$2: \begin{cases} 1: \text{substitution} \\ 1: \text{answer} \end{cases}$
(c) The third-degree Taylor polynomial for f about $x=1$ is $8+3(x-1)-\frac{2}{2}(x-1)^2+\frac{3/2}{3!}(x-1)^3$ $=8+3(x-1)-(x-1)^2+\frac{1}{4}(x-1)^3.$	$2: \begin{cases} 1: \text{two terms} \\ 1: \text{remaining terms} \end{cases}$
(d) $\dfrac{\max\limits_{1\le x\le 1.1}\left\lvert f^{(4)}(x)\right\rvert}{4!}\cdot(1.1-1)^4 \le \frac{300}{4!}\cdot(1.1-1)^4=\frac{300}{24}\cdot 0.1^4=\frac{1}{800}$ An upper bound on the error of the approximation is $\frac{1}{800}$.	$2: \begin{cases} 1: \text{form of the error bound} \\ 1: \text{answer} \end{cases}$